

SARATHI P R

Bengaluru, India (Open to Relocate to Germany)

📞 +91 88929 77046 ✉ sarathijsa@gmail.com  LinkedIn  GitHub

Microsoft Certified: Azure AI Fundamentals (AI-900)

Summary

Machine Learning Engineer with 4+ years of experience in computer vision and deep learning across industrial inspection, surveillance camera analytics, and AR/VR applications. Strong background in end-to-end ML pipelines including data annotation, preprocessing, experimentation, model optimization, and deployment. Experienced in translating research ideas into production-ready models optimized for performance, accuracy, and real-world constraints.

Core Skills

- **Computer Vision:** Object detection, anomaly detection, image classification, segmentation, pose estimation, gesture recognition, OCR
- **Deep Learning:** CNNs, YOLO, Mini VGGNet, SVM, supervised learning, feature extraction
- **Frameworks:** PyTorch, TensorFlow, Keras, OpenCV, scikit-learn
- **Cloud & Certification:** Microsoft Azure AI (AI-900)
- **Edge & Optimization:** Model benchmarking, latency optimization, inference tuning
- **Data & Experimentation:** Data annotation, preprocessing, dataset curation, error analysis
- **Tools & Platforms:** Python, SQL, NumPy, Pandas, Roboflow, Weights & Biases, Jupyter Notebook
- **Visualization & Integration:** Unity (ML integration), Matplotlib, Tableau

Experience

Mantra Softech

August 2024 - Present

Computer Vision Research Engineer

Bengaluru, India

- Designed and evaluated deep learning models for surveillance camera-based inspection and monitoring systems under strict accuracy requirements.
- Experimented with state-of-the-art detection and classification architectures optimized for edge deployment.
- Improved inference stability and reduced error rates by 18–25% through focused retraining, hyperparameter tuning, and dataset refinement.
- Validated and debugged models directly on edge devices, addressing runtime performance and reliability issues.

Viziverse AI (Remote)

April 2023 - July 2024

Machine Learning Engineer

Cambridge, USA

- Researched and deployed computer vision models for headset-less AR/VR applications including pose estimation and gesture recognition.
- Developed end-to-end ML pipelines covering data capture, preprocessing, training, evaluation, and deployment.
- Optimized models and pipelines for speed and efficiency, reducing latency by 15–20%.
- Integrated ML outputs into Unity-based cross-platform applications in collaboration with mobile and product teams.

Visionify.ai

November 2021 - February 2023

Machine Learning Engineer

Westminster, USA

- Delivered computer vision solutions for industrial inspection and defect detection use cases.
- Handled end-to-end workflows including client requirement analysis, data annotation, preprocessing, model training, testing, and deployment.
- Trained anomaly detection and classification models on large-scale manufacturing image and video datasets.
- Standardized annotation and review workflows, reducing turnaround time by 25%.

Internship Experience

Simplify360

Internship

Data Science Intern

India

- Implemented data organization and cleaning techniques for clients including TataSky, contributing to a 40% reduction in cost per resolution.
- Developed sentiment analysis models using NLP techniques, achieving 95% success in targeted marketing campaigns.
- Built and visualized supervised learning models for minute-wise household power consumption data.

Indata.ai

Internship

Machine Learning Intern

India

- Built image processing-based predictive models to assess food quality and estimate pricing.
- Trained Support Vector Machine (SVM) models achieving 93% classification accuracy.
- Performed exploratory data analysis and visualization using Jupyter Notebook and Matplotlib.
- Trained and mentored 5+ junior ML interns and supervised assigned projects.

SourceEdge

Internship

Machine Learning Intern

India

- Developed object detection models using CIFAR-10 dataset with Mini VGGNet architecture.
- Enhanced model accuracy by 5% using YOLO-based training approaches.
- Applied Python-based ML techniques using Keras and scikit-learn for real-time inference.
- Implemented CRUD operations for structured data storage.

Education

Visvesvaraya Technological University (VTU)

2015 - 2019

Bachelor of Engineering in Computer Science

Bengaluru, India

Languages

- English – Full Professional Proficiency (C2)
- German – Intermediate Working Proficiency (B1)